

Arjun Pareek

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Backend engineer in fintech payments, building agentic AI systems on the side — RAG pipelines, multi-agent orchestration, tool use — without frameworks, from scratch. Transitioning fully into AI engineering.

TECHNICAL SKILLS

Languages: Python, C#, SQL, JavaScript, TypeScript

AI / ML: Anthropic API, LangGraph, RAG, Tool Use & Function Calling, Multi-Agent Orchestration, ChromaDB, sentence-transformers

Frameworks: FastAPI, .NET, ASP.NET Core, Streamlit, Node.js, React

Infrastructure: AWS, MySQL, SQLite, Git, Docker, Kafka, Redis

EXPERIENCE

Backend Engineer | *AuroPay, Aurionpro — Payments Processor Team*

06/2025 – Present

- Sole developer on a production BigCommerce payment gateway plugin — owned architecture, OAuth flow, and AWS deployment end-to-end, enabling merchants to accept UPI, card, and net banking payments at checkout.
- Designed a MySQL schema with idempotency guarantees and a webhook audit trail to prevent duplicate charges and support reconciliation.
- Built the Merchant Fraud Management module — schema, full API suite, and role-based access control for fraud case handling.

AI ENGINEERING PROJECTS

Multi-Modal Damage Claim Verification System | *Python, Anthropic API, Pydantic* | GitHub

06/2026

- Placed **14th of 1,773** (top 0.8%) at the HackerRank Orchestrate Hackathon, solo, in 24 hours — up from 234th the prior edition.
- Separated perception from decision-making: a vision-language model for image observation, a deterministic Python layer for all business logic — fully auditable, no LLM calls needed to change rules.
- Enforced structured output via forced Pydantic tool calls; A/B testing showed direct classification beat chain-of-thought by ~10% accuracy at ~46% lower latency.

Multi-Agent Job Application Pipeline | *Python, LangGraph, Anthropic API, GitHub API* | GitHub

06/2026

- Three-agent pipeline (Resume, Job Discovery, Tailoring) with human-in-the-loop approval at every step; pulls live listings from Greenhouse, Lever, and Ashby ATS APIs.
- Solved a LangGraph re-execution bug by splitting approval steps into expensive-work and interrupt-only nodes, preventing repeated side effects on resume.

Agentic Interview Prep System | *Python, Anthropic Tool Use, Tavily, Streamlit* | Live | GitHub

04/2026

- Multi-tool agentic system built on raw Anthropic tool use, no framework — Tavily for live company research, chunked resume embeddings for retrieval.

OTHER PROJECTS

E-Wallet Microservices | *Java, Spring Boot, Eureka, OAuth, Kafka* | GitHub

07/2024

- Microservices e-wallet backend with Kafka for event-driven communication — 40% throughput improvement over synchronous baseline.

RESEARCH

Deep Learning Models for Laryngoscopic Disease Diagnosis | *IEEE CICN 2024*

- Comparative study of YOLOv9, ResNet-50, ResNet-100, and VGG-16 for laryngeal lesion detection; YOLOv9 outperformed all baselines at 95% accuracy.

Multi-Algorithm Pathfinding for Emergency Medical Navigation | *Springer LNNS, ICIVC 2025*

- Smart ambulance dispatch system dynamically selecting from A*, Dijkstra, BHK, and hybrid ABCO* based on real-time traffic conditions.

EDUCATION

D.J. Sanghvi College of Engineering

B.Tech in Computer Science and Engineering | CGPA: 8

09/2021 – 06/2025

D.J. Sanghvi College of Engineering

Minors in AI/ML | GPA: 7.8

09/2023 – 06/2025

ACHIEVEMENTS

Certified in Cybersecurity | (ISC)²

07/2023

Solved 150+ DSA Problems | *LeetCode, HackerRank, GeeksforGeeks*